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ASR Accuracy Metrics: WER, CER, and Their Weaknesses

Word Error Rate (WER) and Character Error Rate (CER)

WER is the standard ASR metric, defined as: [deepgram](#)

$$\text{WER} = \frac{S + D + I}{N}$$

where S = substitutions, D = deletions, I = insertions, and N = total words in reference. WER is computed via minimum edit distance (Levenshtein alignment) between hypothesis and reference transcripts. [deepgram](#)

CER (Character Error Rate) is the character-level analogue, used for languages without clear word boundaries (Chinese, Japanese) or agglutinative languages (Tamil, Turkish). [deepgram](#)

Why WER Penalizes Semantically-Correct Variants

WER has **three fundamental weaknesses** that inflate error scores on semantically correct transcriptions: [deepgram](#)

1. **Orthographic variation:** Different valid spellings of the same word are counted as substitutions. Example: "colour" vs. "color", "ibuprofen" vs. "إيبوبروفين" (Arabic transliteration) [deepgram](#)
1. **Morphological agglutination:** In agglutinative languages (Tamil, Malayalam, Turkish), a single spoken utterance can be validly written with different morpheme boundaries. WER counts different segmentations as full word errors. [deepgram](#)
1. **Code-switching script variance:** In Arabic–English or Persian–English code-switching, English loanwords may appear in Latin script or transliterated into Arabic/Persian script. Both are correct, but WER treats them as completely different tokens. [deepgram](#)

Empirical evidence: A 2026 benchmark on Arabic–Persian code-switching found Kendall's τ correlation of only 0.40 between WER and BERTScore for Persian–English (vs. 0.80 for German–English), indicating WER unfairly penalizes semantically equivalent transliterations. [themoonlight](#)

WER Behavior on Code-Switched Speech

Code-switching (CS) exacerbates WER's weaknesses: [deepgram](#)

ISSUE	EXAMPLE	WER PENALTY
Script variance	"feature" vs. "فیچر" (Persian transliteration)	Full substitution (1.0 WER) despite semantic equivalence themoonlight
Morphological blending	Arabic prefix "ال" + English word ("الfeature")	Counted as insertion + substitution themoonlight
Token alternation	"I need to book a flight *بكره*" (Arabic "tomorrow")	WER assumes single language; penalizes script switches deepgram

Alternative metrics for CS: [deepgram](#)

- **BERTScore:** Token-level cosine similarity in multilingual embedding space (mBERT, XLM-R). Captures semantic equivalence across scripts. [themoonlight](#)

- **Mixed Error Rate (MER):** Combines WER for word-based segments and CER for character-based segments within one transcription [deepgram](#)
- **Script-Aware Error Rate (SAER):** Explicitly penalizes script-switching errors [deepgram](#)
- **Switch Entry Token Error Rate (SETER):** Measures error rate specifically at code-switch boundaries [arxivtldr](#)

BRIDGE framework (Deepgram, 2026): Composite 7-metric evaluation for Indic ASR: [deepgram](#)

- **B:** BERTScore (semantic similarity)
- **R:** Entity F1 (named entity accuracy)
- **I:** Word Information Lost (WIL, normalized complement to WER)
- **D:** Domain-weighted accuracy (AWWER for agricultural/medical terms)
- **G:** Grapheme-level error rate (CER)
- **E:** Error segmentation (SER + MER/SAER for code-switching)

Typical WER Benchmarks

SYSTEM	MONOLINGUAL WER	CODE-SWITCHED WER	NOTES
Whisper Large v3	3–6% (LibriSpeech)	15–25% (Arabic–English) themoonlight	Non-streaming; struggles with CS
ElevenLabs Scribe v2	5–8% (conversational)	13.2% (CS average) themoonlight	Best on Arabic pairs; BERTScore 0.936
Google Chirp 3	6–10% (conversational)	18–22% (CS average) themoonlight	Language-agnostic end-to-end
Deepgram Nova-3	4–7% (English)	N/A (no Arabic/Persian support) themoonlight	Native CS for Latin-script pairs

TTS Quality Metrics: MOS, CMOS, Intelligibility

Mean Opinion Score (MOS)

MOS is the gold standard for TTS quality, asking human raters to score naturalness on a 1–5 scale: [linkedin](#)

$$\text{MOS} = \frac{1}{N} \sum_{i=1}^N r_i$$

where N = number of listeners, r_i = rating from listener i . [convozen](#)

Limitations: [arxiv](#)

- **Expensive:** Requires 20–50 trained raters per condition; not scalable for iterative development
- **Subjective:** Inter-rater variance can be ± 0.5 MOS points
- **Saturation:** At MOS > 4.5, neural MOS predictors cannot reliably differentiate frontier models [linkedin](#)

CMOS (Comparative MOS) and AB Tests

CMOS (Comparative MOS) uses pairwise comparison (System A vs. System B) rather than absolute scoring, reducing rater bias. Results are reported as preference percentages (e.g., "System A preferred 62% of the time"). [convozen](#)

Intelligibility Metrics

Intelligibility measures whether synthesized speech is understandable, distinct from naturalness. Key metrics: [linkedin](#)

METRIC	DESCRIPTION	TARGET
WER (ASR-based)	Run TTS output through ASR; compare transcript to original text	WER < 5% for production linkedin
CER	Character-level intelligibility (for Chinese, Japanese)	CER < 3% linkedin
STOI (Short-Time Objective Intelligibility)	Signal-processing metric predicting speech intelligibility	STOI > 0.85 github.univ-lemans
UTMOSv2	Neural MOS predictor (Pearson correlation 0.84+ with human MOS)	UTMOS > 4.0 for production linkedin

Speaker Similarity and Prosody

METRIC	DESCRIPTION	TARGET
SIM-o (Speaker Similarity)	Cosine similarity of speaker embeddings (WavLM, ECAPA) from reference vs. synthesized audio	SIM-o > 0.85 for voice cloning linkedin
f0 variation	Expressivity score based on pitch contour variation	f0 contour matches reference git-lium.univ-lemans
MCD (Mel Cepstral Distortion)	Euclidean distance between MFCCs of synthesized vs. reference audio	MCD < 5 dB mbrenndoerfer

Automated TTS Evaluators and Their Weaknesses

A 2026 study benchmarked **four MOS predictors** (UTMOSv2, DNSMOS, NISQA, TTSQA) and **four Audio-LLM judges** (Qwen-Audio, Llama-3-Audio, etc.) on 860 linguistically annotated utterances. [arxiv](#)

Key findings: [arxiv](#)

- **MOS predictors collapse onto acoustic signal quality:** They detect noise, distortion, and artifacts but miss linguistically structured errors (mispronunciation, wrong stress, unnatural prosody)
- **Audio-LLM judges show selective, prompt-dependent detection:** They can detect specific error types (e.g., "check for mispronounced words") but do not generalize across all 10 perceptual dimensions
- **Neither class reliably captures breadth of errors:** Automated evaluators miss ~40% of linguist-annotated errors, particularly prosodic and morphological mistakes

Recommendation: Use automated MOS predictors for rapid iteration, but validate with human MOS for production releases. [arxiv](#)

Latency Metrics: Inconsistent TTFA Definitions

Time-to-First-Audio (TTFA) Definition Variance

TTFA (Time-to-First-Audio) measures latency from TTS request submission to first audio sample arrival. However, vendors define TTFA inconsistently: [bland](#)

VENDOR	CLOCK START	CLOCK STOP	NOTES
Coval	Synthesis start (after connection established)	First audio chunk arrival + leading silence inside stream	TTFA = (first audio chunk arrival - synthesis start) + leading silence gradium
Bland AI	After connection established, on warm server, with short input sentence	First byte of audio (not first intelligible word)	Vendors control where clock starts/stops; end-to-end window longer than isolated TTFA bland
ElevenLabs	Text submission to API	First audio byte received	Does not include network handshake or LLM inference
Gradium	LLM generates first token	First audible sample (excluding leading silence)	Median TTFA 214 ms (Sept 2026) gradium

Why inconsistency matters: A vendor reporting 100 ms TTFA (warm connection, short sentence) may actually deliver 400 ms end-to-end in production (cold start, long sentence, network latency). [bland](#)

End-to-End Latency Budget

Production voice agents must account for **full pipeline latency**, not just TTS TTFA: [bland](#)

STAGE	TYPICAL LATENCY (P50)	NOTES
VAD end-of-speech detection	20–100 ms	Semantic VAD reduces false endpoints
ASR streaming first partial	100–300 ms	Streaming; final transcript adds 50–100 ms
LLM time-to-first-token (TTFT)	200–600 ms	Dominates pipeline; depends on model size
TTS time-to-first-audio (TTFA)	75–300 ms	Modern APIs (e.g., Realtime TTS-2 Flash) achieve <100 ms P99

STAGE	TYPICAL LATENCY (P50)	NOTES
Network + orchestration	50–200 ms	WebRTC <50 ms; PSTN 150–700 ms
Total TTFA	600–1,200 ms	Sub-500 ms achievable with aggressive streaming

Recommended Latency Metrics

METRIC	DEFINITION	TARGET
TTFA (isolated)	TTS request → first audio byte	<150 ms (warm)
End-to-end TTFA	User stops speaking → agent starts speaking	<500 ms (P50), <1,000 ms (P95) softwareseni
RTF (Real-Time Factor)	Inference time / audio duration	RTF < 1.0 (faster than real-time) linkedin
Turn gap	User utterance end → agent utterance start	160–230 ms (human parity)

End-to-End Conversational Quality Evaluation

LLM-as-Judge Approaches

LLM-as-judge uses large language models to evaluate voice agent conversations holistically, scoring dimensions like helpfulness, naturalness, and task completion. [arxiv](#)

Typical setup: [arxiv](#)

1. Record full conversation (user audio + agent audio + transcripts)
2. Prompt LLM (e.g., GPT-4o, Claude) with conversation transcript + evaluation rubric
3. LLM scores each dimension on 1–5 scale with justification

Advantages: [arxiv](#)

- **Scalable:** No human raters needed; can evaluate thousands of conversations

- **Multi-dimensional:** Can score helpfulness, naturalness, empathy, task completion simultaneously
- **Explainable:** LLM provides rationale for each score

Weaknesses: [arxiv](#)

- **Prompt-dependent:** Scores vary significantly based on prompt wording and few-shot examples
- **Bias toward verbosity:** LLMs prefer longer, more detailed responses, penalizing concise agents
- **Cannot perceive audio quality:** LLM judges text transcripts only; misses prosody, emotion, and acoustic artifacts

Full-Duplex-Bench and Turn-Taking Metrics

Full-Duplex-Bench v1.5 (NTU + UW + Berkeley + CMU, 2026) is an automated benchmark for overlap handling in full-duplex voice agents.

Metrics:

- **Stop latency:** Time from user interruption onset to agent stopping speech
- **Response appropriateness:** Whether agent correctly identifies overlap type (backchannel, barge-in, background speech)
- **Context recovery:** Whether agent resumes conversation coherently after interruption

Turn-Taking metrics (Gadium, 2026):

- **Interruption rate:** % of user utterances where agent interrupts mid-sentence
- **Backchannel detection accuracy:** % of backchannels correctly identified (not triggering turn end)
- **Turn gap distribution:** Histogram of user-end → agent-start intervals (target: 160–230 ms)

Why Voice Agent Evaluation Is Harder Than Text Agent Evaluation

Voice agent evaluation is fundamentally more complex than text agent evaluation due to **four additional dimensions:** [arxiv](#)

DIMENSION	TEXT AGENT	VOICE AGENT	WHY HARDER
Timing	Irrelevant (asynchronous)	Critical (latency, turn gap)	Must measure P50/P95 latency across regions; human parity requires 160–230 ms turn gaps
Audio quality	N/A	MOS, intelligibility, speaker similarity	Requires human raters or neural MOS predictors; automated evaluators miss 40% of errors arxiv
Multi-modal signals	Text only	Audio + text + timing + prosody	Must evaluate backchannels, interruptions, emotion; LLM-as-judge cannot perceive audio arxiv
Subjective quality	Helpfulness, correctness	Helpfulness + naturalness + empathy + prosody	MOS saturation at 4.5; requires fine-grained linguistic annotation (10 dimensions) arxiv

Additional challenges: [deepgram](#)

- **Code-switching:** WER unfairly penalizes semantically correct transliterations; requires BERTScore + MER/SAER [themoonlight](#)
- **Dialect variation:** Same language (e.g., Arabic) has 30+ dialects; WER varies 10–20% across dialects [themoonlight](#)
- **Domain terminology:** Headline WER can hide 2× higher error rates on domain-critical terms (medical, agricultural) [deepgram](#)

Key Papers and Benchmarks

PAPER / BENCHMARK	YEAR	CONTRIBUTION
Beyond Naturalness (Bamgbose et al., arXiv:2608.09930)	2026	Deconstructs TTS naturalness into 10 linguistic dimensions; benchmarks MOS predictors and Audio-LLM judges arxiv
Benchmarking Commercial ASR on Code-Switching (Moonlight, 2026)	2026	1,200 CS utterances (Arabic–English, Persian–English, German–English); argues for BERTScore over WER themoonlight
Why WER Fails Indian Languages (Deepgram, 2026)	2026	BRIDGE 7-metric framework for Indic ASR (BERTScore, Entity F1, WIL, AWWER, CER, SER,

PAPER / BENCHMARK	YEAR	CONTRIBUTION
		MER/SAER) deepgram
Full-Duplex-Bench v1.5 (Lin et al., ICASSP 2026)	2026	Automated benchmark for overlap handling; decomposes interruptions, backchannels, background speech
Beyond Word Error Rate (arXiv:2609.11786)	2026	Evaluates ASR on English–Yoruba CS; proposes switch-localized diagnostics (SETER, windowed switch point error) arxivtldr
TTS MOS Playbook (DupDub, 2025)	2025	Comprehensive guide to TTS evaluation (WER/CER, UTMOSv2, SIM-o, RTF) linkedin

Recommended evaluation stack for production voice agents (2026): [deepgram](#)

- **ASR:** WER + CER + BERTScore + Entity F1 (for code-switching: MER/SAER)
- **TTS:** UTMOSv2 (naturalness) + WER/CER (intelligibility) + SIM-o (speaker similarity)
- **Latency:** End-to-end TTFA (P50/P95) + turn gap distribution
- **Conversational quality:** LLM-as-judge (helpfulness, task completion) + Full-Duplex-Bench (overlap handling)

Voice agent evaluation requires **multi-metric, multi-dimensional** approaches that capture both surface accuracy (WER, MOS) and deeper qualities (semantic fidelity, naturalness, turn-taking timing). Single-number metrics are insufficient for production deployments. [arxiv](#)